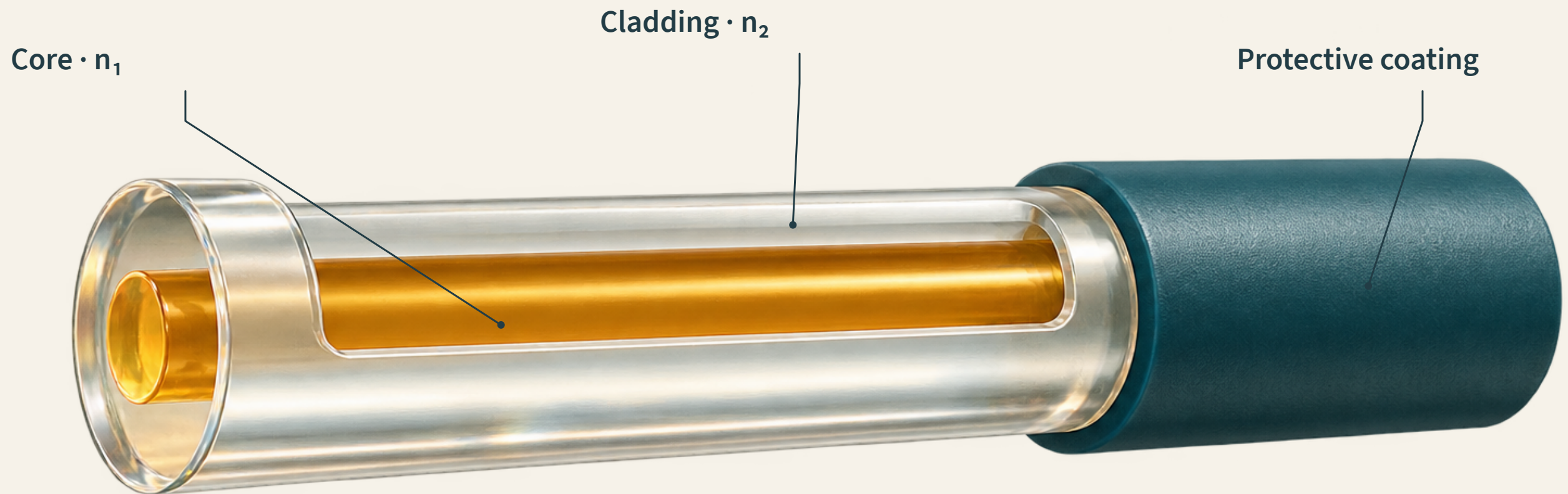
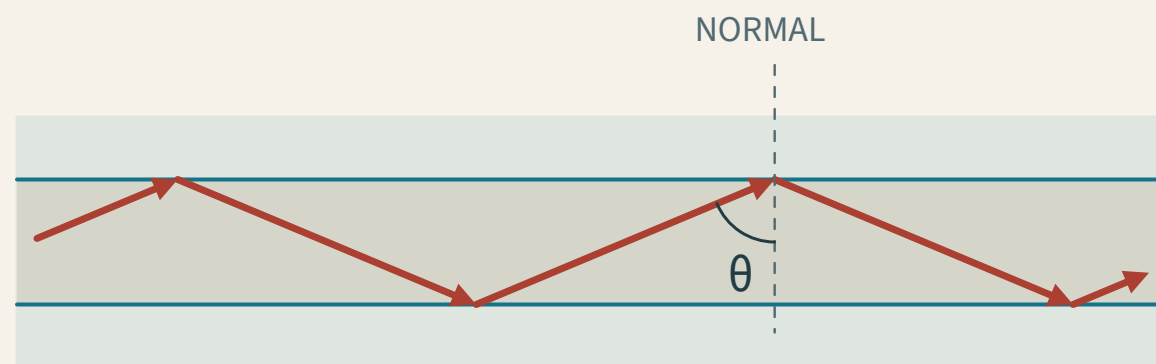


How an optical fibre guides light

One explanation, three visual directions. A conceptual step-index fibre in the ray model.



THE MATERIAL



$$n_1 > n_2$$

Total internal reflection occurs at the core-cladding boundary.

THE OPTICAL CONDITION

$$\theta > \theta_c$$

$$\sin \theta_c = n_2 / n_1$$

θ is measured from the normal.

The coating protects the glass; index contrast guides the light.